

Raw experimental data

一、Overall Results

skew = [0.1, 0.3, 0.5, 0.7, 0.9]

YCSB

serial = [1825, 1806, 1817, 1770, 1719]

pdcc = [19016, 18903, 18693, 16178, 7780]

aria = [17321, 17028, 17163, 11592, 1385]

cm-exe= [15066, 15240, 14612, 7764, 3502]

cm-rep= [19194, 19189, 18880, 15236, 5256]

SMALLBANK

serial = [3535, 3570, 3580, 3589, 3535]

pdcc = [35678, 35784, 34880, 24845, 12105]

aria = [29879, 29910, 29640, 25438, 7461]

cm-exe= [25374, 25595, 24589, 19205, 10384]

cm-rep= [30560, 30432, 29708, 25261, 18613]

二、Scalability

cores = [4, 8, 16, 32]

YCSB_Uniform

serial = [1653, 1618, 1633, 1740]

pdcc = [2787, 6682, 12160, 19053]

aria = [3099, 6556, 11172, 17206]

cm-exe= [3239, 7205, 11290, 15355]

cm-rep= [4115, 7932, 12667, 19451]

YCSB_Zipfian

serial = [1566, 1534, 1500, 1621]

pdcc = [1972, 3766, 6013, 7952]

aria = [209, 442, 859, 1418]

cm-exe= [2338, 3097, 3035, 3267]

cm-rep= [2667, 4288, 4517, 5233]

SMALLBANK_Uniform

serial = [3270, 3157, 3189, 3365]

pdcc = [5856, 13244, 23074, 35549]

```
# aria = [6256, 12557, 20999, 29797]
# cm-exe= [6207, 12393, 19384, 25874]
# cm-rep= [8056, 14781, 22520, 30072]
```

```
# SMALLBANK_Zipfian
serial = [3317, 3196, 3148, 3380]
pdcc = [3601, 6133, 9598, 11910]
aria = [1573, 3243, 5297, 7308]
cm-exe= [4702, 8034, 9229, 10184]
cm-rep= [6391, 11896, 15406, 18567]
```

三、Performance Breakdown

```
labels = ['1', '2', '3', '4', '5', '6', '7', '8']
# YCSB Uniform
data = [0.014, 46.218, 0.815, 3.198, 1.502, 0.009, 3.678, 0.170]
# YCSB Zipfian
data = [0.035, 114.807, 1.165, 4.551, 2.119, 0.227, 5.238, 0.697]
```

四、Re-Validation Ablation Study

```
cores = [4, 8, 16, 32]

# YCSB
pdcc = [1972, 3766, 6013, 7952]
pdcc_without_revalidation = [1956, 3656, 5417, 6042]

# SMALLBANK
# pdcc = [3601, 6133, 9598, 11910]
# pdcc_without_revalidation = [3456, 5848, 9365, 11602]
```

五、Dynamic Batching Effectiveness

```
# Impact of Batch Size on Throughput
skew = [0.1, 0.3, 0.5, 0.7, 0.9]

# YCSB
pdcc = [19185, 18970, 18411, 16411, 7828]
pdcc10 = [19325, 19341, 18881, 16616, 4521]
pdcc5 = [18275, 17845, 17908, 16364, 5907]
pdcc1 = [12018, 12171, 12127, 11814, 7817]
```

```
# SMALLBANK
# pdcc = [34560, 34466, 34057, 25726, 12806]
# pdcc10 = [35007, 34387, 34010, 25032, 7443]
# pdcc5 = [32301, 32430, 32195, 26459, 8194]
# pdcc1 = [21560, 21619, 21583, 21412, 12648]
```

```
# Execution Deviation and Threshold Selection
skew = [0.1, 0.3, 0.5, 0.7, 0.9]
```

```
# YCSB
# pdcc10 = [0.02, 0.02, 0.12, 1.71, 24.35]
# pdcc5 = [0.03, 0.03, 0.08, 1.32, 22.34]
# pdcc1 = [0.00, 0.00, 0.00, 0.77, 18.22]
```

```
# SMALLBANK
pdcc10 = [0.02, 0.03, 0.46, 10.32, 57.52]
pdcc5 = [0.02, 0.03, 0.11, 6.84, 54.99]
pdcc1 = [0.00, 0.00, 0.00, 0.17, 29.90]
```